

CS 506: Introduction to Quantum Computing

Controlled Gates and Quantum Superdense Coding

Scribe: Marco Fabiani

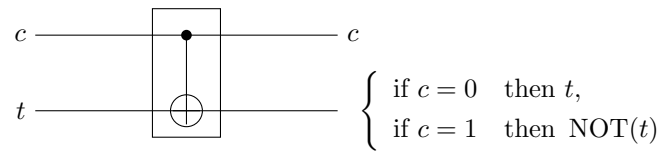
September 21, 2026

1 Overview

This lecture introduces controlled quantum gates, beginning with the C-NOT gate and its generalization to controlled- U operations. It then discusses the Hadamard and $\pi/8$ phase gates and their role in universal gate sets. The lecture next presents quantum superdense coding, showing how two classical bits can be communicated by sending one qubit when Alice and Bob share an entangled state. Finally, it introduces the problem of sending a qubit through a classical channel, motivating quantum teleportation.

2 Control-NOT (C-NOT) 2-Qubit Gate

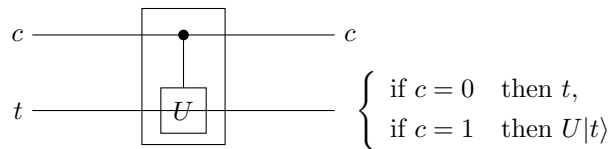
The C-NOT gate is a two-qubit gate with a control qubit c and a target qubit t .



Concisely, the target output is

$$t = c \oplus t.$$

Controlled- U gate. Let U be any 2×2 unitary operator. The corresponding controlled- U , or $C-U$, gate is



Its action is

$$C-U |0\rangle|t\rangle = |0\rangle|t\rangle,$$

$$C-U |1\rangle|t\rangle = |1\rangle(U|t\rangle).$$

C- U gate in Dirac notation. The controlled- U operator can be written as

$$C-U = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U,$$

where $\otimes$ denotes the tensor product and I denotes the 2×2 identity.

For control state $|0\rangle$,

$$C-U |0\rangle|t\rangle = |0\rangle\langle 0|0\rangle \otimes I|t\rangle + |1\rangle\langle 1|0\rangle \otimes U|t\rangle = |0\rangle|t\rangle,$$

since $\langle 0|0\rangle = 1$ and $\langle 1|0\rangle = 0$.

For control state $|1\rangle$,

$$C-U |1\rangle|t\rangle = |0\rangle\langle 0|1\rangle \otimes I + |1\rangle\langle 1|1\rangle \otimes U|t\rangle = |1\rangle U|t\rangle,$$

since $\langle 0|1\rangle = 0$ and $\langle 1|1\rangle = 1$.

C-NOT as a controlled- U gate. If

$$U \equiv \text{NOT} = |0\rangle\langle 1| + |1\rangle\langle 0|,$$

then

$$C\text{-NOT} \equiv |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes (|0\rangle\langle 1| + |1\rangle\langle 0|)$$

and therefore

$$C\text{-NOT} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes |0\rangle\langle 1| + |1\rangle\langle 1| \otimes |1\rangle\langle 0|.$$

Hadamard gate. For $U \equiv H$,

$$H|0\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = |+\rangle,$$

$$H|1\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle = |-\rangle.$$

The corresponding controlled- H gate satisfies

$$C|0\rangle|t\rangle = |0\rangle|t\rangle,$$

$$C|1\rangle|t\rangle = |1\rangle(H|t\rangle).$$

$\pi/8$ phase gate. The one-qubit gate T is

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix} = e^{i\pi/8} \begin{pmatrix} e^{-i\pi/8} & 0 \\ 0 & e^{i\pi/8} \end{pmatrix},$$

where $e^{i\pi/8}$ is the global phase.

Its action on the computational basis is

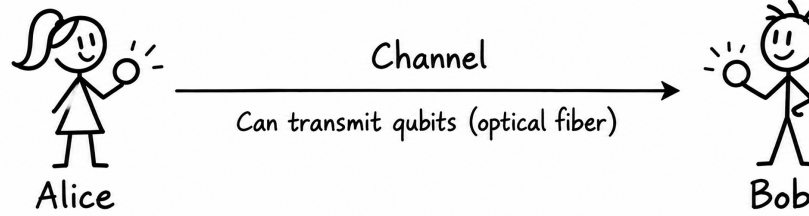
$$\begin{cases} T|0\rangle = |0\rangle, \\ T|1\rangle = e^{i\pi/4}|1\rangle. \end{cases}$$

Universal gates. The following properties were stated:

- Any N -qubit gate can be realized using H , CNOT, and T gates.
- Any 1-qubit gate can be realized using H and T gates, or R_x , R_y , and R_z gates.

3 Quantum Superdense Coding

Alice and Bob are connected by a channel that can transmit qubits.



Alice wants to send 2 classical bits to Bob: 00, 01, 10, or 11. The question is whether Alice can communicate these two classical bits by sending only one qubit.

The following one-qubit gates will be used:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I|0\rangle = |0\rangle, \quad I|1\rangle = |1\rangle,$$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle,$$

and

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

Moreover, $I^2 = X^2 = Z^2 = I$.

Shared two-qubit state. Before going away, Alice and Bob share the following entangled state:

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}} \underbrace{|0\rangle}_{\text{Alice's qubit}} \otimes \underbrace{|0\rangle}_{\text{Bob's qubit}} + \frac{1}{\sqrt{2}} \underbrace{|1\rangle}_{\text{Alice's qubit}} \otimes \underbrace{|1\rangle}_{\text{Bob's qubit}}.$$

The first qubit belongs to Alice and the second qubit belongs to Bob. The two qubits are entangled.

Sending 2 classical bits. The protocol proceeds as follows:

1. Alice performs an operation on her qubit.
2. Alice sends her qubit to Bob.
3. Bob measures the two qubits in the Bell basis

$$(|\beta_{00}\rangle, |\beta_{01}\rangle, |\beta_{10}\rangle, |\beta_{11}\rangle)$$

and recovers the classical bits Alice wanted to send.

This provides a very secure communication scheme.

The four possible cases are shown on the following page.

The four possible encodings are summarized below.

Alice wants to send	Alice applies this gate on her qubit	New state of the two qubits	Alice sends her qubit	Bob measures both qubits
00	Nothing (apply I gate)	$ \beta_{00}\rangle = \frac{1}{\sqrt{2}} 00\rangle + \frac{1}{\sqrt{2}} 11\rangle$	$\longrightarrow$	$ \beta_{00}\rangle$ with probability 1
01	Apply X	$\frac{1}{\sqrt{2}}(X 0\rangle) 0\rangle + \frac{1}{\sqrt{2}}(X 1\rangle) 1\rangle$ $= \frac{1}{\sqrt{2}} 1\rangle 0\rangle + \frac{1}{\sqrt{2}} 0\rangle 1\rangle$ $= \beta_{01}\rangle$	$\longrightarrow$	$ \beta_{01}\rangle$ with probability 1
10	Apply Z	$\frac{1}{\sqrt{2}}(Z 0\rangle) 0\rangle + \frac{1}{\sqrt{2}}(Z 1\rangle) 1\rangle$ $= \frac{1}{\sqrt{2}} 00\rangle - \frac{1}{\sqrt{2}} 11\rangle$ $= \beta_{10}\rangle$	$\longrightarrow$	$ \beta_{10}\rangle$ with probability 1
11	Apply X , then apply Z $\Rightarrow (Z \cdot X)$	$\frac{1}{\sqrt{2}}(ZX 0\rangle) 0\rangle + \frac{1}{\sqrt{2}}(ZX 1\rangle) 1\rangle$ $= \frac{1}{\sqrt{2}}(Z 1\rangle) 0\rangle + \frac{1}{\sqrt{2}}(Z 0\rangle) 1\rangle$ $= -\frac{1}{\sqrt{2}} 10\rangle + \frac{1}{\sqrt{2}} 01\rangle$ $= \frac{1}{\sqrt{2}} 01\rangle - \frac{1}{\sqrt{2}} 10\rangle$ $= \beta_{11}\rangle$	$\longrightarrow$	$ \beta_{11}\rangle$ with probability 1

4 Sending a Qubit over a Classical Channel

Suppose Alice wants to send a qubit to Bob, but the communication channel can transmit only classical bits. Consider, for example, the state

$$|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle.$$

The first amplitude can be written numerically as

$$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} = \frac{1.732\dots}{3} = 0.577\dots$$

The decimal expansion does not terminate, so sending only a finite numerical approximation would not specify this amplitude exactly.

This motivates the question: how can Alice send the qubit to Bob exactly if the channel accepts only classical bits?

This leads to the next topic: **quantum teleportation**.